

Dashboard Chart Guide For Real-Time Operation

This file explains each dashboard chart/panel, what it means in real time, and how a real grid operator would use it.

Dashboard file:

- `frontend/src/App.jsx`

API client:

- `frontend/src/api/client.js`

Refresh Behavior

Mode	Refresh frequency	Data source
Normal API mode	30 seconds	Backend historical/processed EMS data.
Live simulation mode	1.5 seconds	Backend live telemetry simulator.
Offline cache mode	Static snapshot	<code>frontend/public/api-cache/*.json</code> .

When the operator clicks **Start live stream**, the frontend calls:

```
POST /live/start
```

Then it polls the backend every 1.5 seconds.

Top Command Bar

The top bar shows:

- Connection state: online, refreshing, degraded, or offline cached snapshot.
- Current refresh cadence: 1.5s live tick or 30s polling.
- Last updated time.
- Active API base URL.
- Start/stop live stream button.
- Manual refresh button.

Operator use:

- Confirms whether data is live or cached.
- Prevents accidental interpretation of stale data as live data.
- Gives immediate visibility into backend connectivity.

KPI Strip

The dashboard shows six key values at the top.

1. Solar

Meaning:

- Current PV generation in kW.

How operator uses it:

- Checks whether renewable supply is available.
- Low solar during daytime can indicate cloud cover or renewable drop.
- Zero solar at night is expected.

2. Load

Meaning:

- Current facility demand in kW.

How operator uses it:

- Detects morning/evening peaks.
- Watches for demand spikes.
- Compares load against solar and battery output.

3. BESS SoC

Meaning:

- Battery state of charge in percent.

How operator uses it:

- Confirms battery is inside 20 to 90 percent safe operating band.
- Low SoC near evening peak means limited peak support.
- High SoC before evening peak is useful because stored energy can offset expensive grid import.

4. Grid

Meaning:

- Current grid import power in kW.

How operator uses it:

- High grid import during peak tariff means cost risk.
- If grid import drops during battery discharge, dispatch is working.
- In island mode, grid import should go to zero and load shedding may appear if supply is insufficient.

5. Tariff

Meaning:

- Current grid import cost in INR/kWh.

How operator uses it:

- Off-peak: charging may be acceptable.
- Mid tariff: usually hold unless solar surplus exists.
- Peak tariff: discharge battery if SoC allows.

6. Savings

Meaning:

- Estimated percentage savings against baseline grid import cost.

How operator uses it:

- Measures whether EMS dispatch is economically useful.
- Can be used as a management KPI in reports.

Power Flow Chart

Signals shown:

- Load kW.
- Solar kW.
- Grid kW.
- Battery SoC percent.

What it tells:

- Whether load is being served by solar, grid, or battery.
- Whether battery SoC is rising or falling.
- Whether grid import increases during solar drop or load rise.

Operator interpretation:

Pattern	Meaning
Solar above load and SoC rising	Battery is storing renewable surplus.
Load above solar and grid high	Facility depends on grid.
Peak tariff and grid falling	Battery discharge is reducing cost.
SoC stuck near 20 percent	Battery reserve is low.
SoC near 90 percent with solar surplus	Battery is close to full, curtailment may occur.

Dispatch Panel

Shows:

- Current recommended action.
- Explanation/reason.
- Override buttons.

Actions:

Action	Meaning
Charge	Put energy into BESS.
Discharge	Use BESS to serve load or reduce grid import.
Idle	Keep battery unchanged.

Override modes:

Mode	What it does
Auto dispatch	Backend dispatch logic decides.
Force charge	Pushes the battery toward charging within safe limits.
Force discharge	Pushes the battery toward discharging within safe limits.
Island mode	Sets grid import to zero and uses solar/battery only; may create load shedding if supply is not enough.

Operator use:

- Understands not only what the EMS recommends, but why.
- Tests emergency behavior.
- Demonstrates operator authority over AI recommendations.

Forecast Chart

Signals shown:

- Forecast solar kW.
- Forecast load kW.

What it tells:

- Expected renewable availability.
- Expected demand.
- Whether future deficit or surplus is likely.

Operator use:

- Prepare for evening peak.
- Pre-charge battery before high tariff.
- Identify low solar periods.
- Avoid using battery too early if future high tariff/load is expected.

Alerts Panel

Alert types:

Alert	Meaning
<code>peak_demand_risk</code>	Load is near or above high-load threshold.
<code>renewable_drop</code>	Solar is expected to fall sharply.
<code>battery_low_soc</code>	Battery is near minimum safe SoC.
<code>battery_high_soc</code>	Battery is near maximum safe SoC.
<code>peak_tariff_import</code>	Grid import is happening during expensive tariff.

Operator use:

- Prioritizes attention.
- Supports fast response.
- Gives concrete messages rather than raw data only.

Cost Chart

Shows:

- Baseline cost.
- Optimized cost.

Baseline means:

```
grid supplies load after solar, with no intelligent battery dispatch
```

Optimized means:

```
dispatch engine uses battery according to tariff, solar surplus, and SoC constraints
```

Operator use:

- Demonstrates economic benefit.
- Supports energy management reporting.
- Useful for monthly/weekly savings communication.

Sustainability Chart

Shows:

- Renewable share.
- Grid dependency.
- Self-sufficiency.

Operator use:

- Tracks carbon and sustainability goals.
- Shows whether the microgrid is using local renewable energy effectively.
- Helps justify battery and PV investments beyond cost savings.

Scenario Panel

Available scenarios:

Scenario	What changes
Normal	Uses current dataset/window.
Peak load	Increases load by 24 percent within feeder limits.
Low solar	Reduces daylight PV output to 38 percent.
Tariff spike	Raises peak tariff to 11.5 INR/kWh.

Operator use:

- Tests system robustness before an event.
- Shows how cost, grid dependency, alerts, and dispatch decisions change.
- Good for viva/demo because it proves the dashboard is interactive.

Dispatch Schedule Table

Columns:

Column	Meaning
Time	Timestamp of decision row.
Solar	PV output in kW.
Load	Facility demand in kW.
SoC	Battery charge level.
Grid	Required grid import.
Action	Charge, discharge, or idle.
Reason	Human-readable dispatch explanation.

Operator use:

- Provides auditability.
- Shows not only the current action but near-term planned behavior.
- Helps explain decisions to supervisors or reviewers.

How To Tell If Dashboard Is Live

The dashboard is live if:

- Top bar says **online**.

- It shows **1.5s live tick**.
- Live records count increases.
- The latest update time changes every few seconds.
- **GET /live/status** returns **running: true**.

If it says **offline: cached snapshot**, then the backend is not reachable and the data is static.